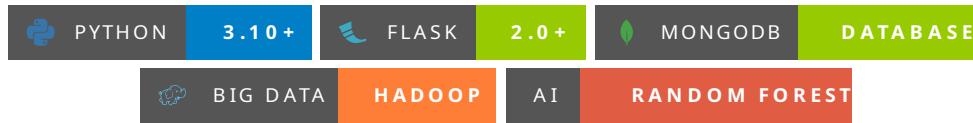


EarthScape Climate Monitoring System



EarthScape is a comprehensive Big Data solution designed to monitor global temperature trends, ingest satellite data, and predict future climate anomalies using Machine Learning.

Overview

Climate change requires robust data processing capabilities. EarthScape provides a unified platform for:

1. **Ingesting** massive datasets from weather stations and satellites into **HDFS**.
2. **Processing** historical records using **MapReduce** algorithms.
3. **Predicting** future temperatures with a **Random Forest** regression model.
4. **Visualizing** trends via an interactive Analyst Dashboard.

System Architecture

The application follows a modular MVC (Model-View-Controller) pattern integrated with Big Data pipelines.

1. **Frontend:** Flask (Jinja2 Templates), Bootstrap 5.
2. **Backend:** Python 3.10+.
3. **Database:** MongoDB (User credentials & logs).
4. **Storage:** Hadoop Distributed File System (HDFS) for raw CSVs.
5. **Analytics:** Scikit-Learn (ML) and Pandas.

Tech Stack

Component

Web Framework

Database

Big Data

ML Libraries

Visualization

Frontend

Technology

Flask

MongoDB Community

Apache Hadoop, HDFS, MapReduce

Scikit-Learn, Pandas, NumPy

Matplotlib, Seaborn

HTML5, CSS3, Bootstrap 5, Bootstrap Icons

Installation & Setup

Prerequisites

- Python 3.10+
- MongoDB (Running locally on default port 27017)
- Git LFS (Required to download the model files)

1. Clone the Repository

Ensure you have Git LFS installed to pull the large model files automatically.

```
git lfs install
git clone https://github.com/MohsinKhanAptech/EarthScape-Climate-Agency.git
cd earthscape-climate-system
```

2. Navigate to Application Code

The source code is located in the website directory.

```
cd EarthScape-Website
```

3. Create & Activate Virtual Environment

```
# Windows
python -m venv .env
.venv\Scripts\activate

# Mac/Linux
python3 -m venv .env
source .venv/bin/activate
```

4. Install Dependencies

```
pip install -r requirements.txt
```

5. Configure Environment

Create a .env file in the EarthScape-Website folder:

```
MONGO_URI=mongodb://localhost:27017/earthscape_db
SECRET_KEY=your_dev_secret_key
MAIL_SERVER=smtp.gmail.com
MAIL_PORT=587
MAIL_USE_TLS=True
MAIL_USERNAME=your-email@example.com
MAIL_PASSWORD=your-app-password
```

6. Run the Application

```
python run.py
```

Visit <http://127.0.0.1:5000> in your browser.

Optional: Retraining the Model

If you want to update the dataset or regenerate the models yourself:

```
python training/train_model.py
```

Usage Guide

1. The Analyst Dashboard

1. **Sign Up** as a new user.
2. Navigate to the **Dashboard**.
3. Select a **City** (e.g., London) and a **Future Date** (e.g., July 2030).
4. Click **Run Prediction**.
5. View the predicted temperature and the generated trend line graph.

2. Data Ingestion (Admin)

1. Navigate to the **Ingestion** tab.
2. Upload a **.csv** dataset.
3. The system will simulate an upload to the `/user/earthscape/climate_data/` HDFS directory.

3. Running Big Data Scripts (Terminal)

To run the MapReduce job manually:

```
python big_data/map_reduce_jobs.py data/GlobalLandTemperaturesByMajorCity.csv
```

4. Retraining the Model (Optional)

If you update the dataset or want to regenerate the `.pkl` files, run this from the EarthScape-Website directory:

```
python training/train_model.py
```

Project Structure

```
EarthScape-Climate-Agency/
├── Dataset/           # Raw Global Climate CSV Data
├── Docs/              # Documentation, Reports & Diagrams
├── EarthScape-Website/ # Main Application Source Code
│   ├── app/
│   │   ├── models/    # ML Models (.pkl) & Logic
│   │   ├── static/    # CSS, Images, Generated Plots
│   │   ├── templates/ # HTML Jinja2 Templates
│   │   └── ...
│   ├── big_data/      # Hadoop & MapReduce Scripts
│   ├── training/      # Model Training Scripts (train_model.py)
│   ├── config/        # Configuration Classes
│   ├── tests/         # Unit Tests
│   ├── run.py         # Application Entry Point
│   ├── requirements.txt # Python Dependencies
└── README.md         # Project Documentation
```

Contributing

1. Fork the repository.
2. Create your feature branch (git checkout -b feature/AmazingFeature).
3. Commit your changes (git commit -m 'Add some AmazingFeature').
4. Push to the branch (git push origin feature/AmazingFeature).
5. Open a Pull Request.

License

This project is licensed under the MIT License.

Author

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